FURTHER MECHANICS / WORK, ENERGY AND POWER

Power and Driving Force

Worked solutions 12 questions · 107 marks

EXAM BOARDS Edexcel FM1 9FM0/3C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Minor Y431 OCR MEI Mech Major Y421

Instructions

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves g unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

Questions

Question 1	*8 marks*	2
Question 2	*5 marks*	4
Question 3	*14 marks*	5
Question 4	*11 marks*	7
Question 5	*7 marks*	9
Question 6	*10 marks*	11
Question 7	*7 marks*	13
Question 8	*8 marks*	15
Question 9	*9 marks*	17
Question 10	*9 marks*	19
Question 11	*8 marks*	21
Question 12	*11 marks*	23

1. A maintenance train has mass 10 000 kg and the maximum power of its engine is 200 kW. In a model of the train's motion, it is assumed that while the train is moving the total resistance to motion is constant.

On a straight horizontal track, the greatest constant speed that the train can maintain when the engine is working at maximum power is $20\,\mathrm{m\,s^{-1}}$.

(a) Find the resistance to motion. **[2]**

(b) While the train is still on horizontal track, the engine is delivering 60 kW when the speed of the train is $8\,\mathrm{m\,s^{-1}}$.
Find the acceleration of the train at this instant. **[2]**

(c) The train then climbs a straight track inclined at an angle θ above the horizontal. When the engine is working at maximum power, the greatest constant speed that the train can maintain is $10\,\mathrm{m\,s^{-1}}$.
Determine the value of θ. **[4]**

Solution

(a) At the greatest constant speed on horizontal track, the train has zero acceleration, so the resultant force is zero. Hence the driving force F equals the resistance, say R, so $F = R$.

With driving force F at speed v, the power is

$$P = Fv$$

Substituting $P = 200\,000$, $F = R$ and $v = 20$:

$$200\,000 = R \times 20$$

so

$$R = 10\,000\,\mathrm{N}$$

(b) When the power is 60 kW and the speed is $8\,\mathrm{m\,s^{-1}}$, the driving force is

$$F = \frac{P}{v} = \frac{60\,000}{8} = 7500\,\mathrm{N}$$

Take the direction of motion as positive. The resultant force is

$$7500 - 10\,000 = -2500\,\mathrm{N}$$

Using Newton's second law:

$$a = \frac{-2500}{10\,000} = -0.25\,\mathrm{m\,s^{-2}}$$

(c) On the incline, at the greatest constant speed $10\,\mathrm{m\,s^{-1}}$, the train again has zero acceleration, so the driving force balances the resistance and the component of weight down the slope.

At maximum power:

$$F = \frac{P}{v} = \frac{200\,000}{10} = 20\,000\,\mathrm{N}$$

The forces opposing the motion are

$$R + mg\sin\theta = 10\,000 + 10\,000 \times 9.8\sin\theta$$

So

$$20\,000 = 10\,000 + 10\,000 \times 9.8 \sin\theta$$

$$10\,000 = 98\,000 \sin\theta$$

$$\sin\theta = \frac{10\,000}{98\,000} = \frac{5}{49}$$

Therefore

$$\theta = \sin^{-1}\left(\frac{5}{49}\right) \approx 5.86^\circ$$

Final answers with $g = 10$: (c) $\sin^{-1}(0.1) \approx 5.74^\circ$

2. A lorry of mass 2400 kg is towing a trailer of mass 600 kg up a straight road inclined at an angle θ to the horizontal, where $\sin\theta = \frac{1}{49}$.

When the lorry and trailer move at speed $v\,\mathrm{m\,s^{-1}}$, the total resistance to motion has magnitude $(300 + 60v)\,\mathrm{N}$.

The greatest constant speed that the lorry and trailer can maintain on this hill is $15\,\mathrm{m\,s^{-1}}$.

Find the maximum power output of the lorry's engine.

Fully justify your answer. **[5]**

Solution

For the lorry and trailer together, the total mass is

$$2400 + 600 = 3000\,\mathrm{kg}$$

The component of the total weight acting down the slope is

$$3000g\sin\theta = 3000 \times 9.8 \times \frac{1}{49} = 600\,\mathrm{N}$$

When the speed is $v\,\mathrm{m\,s^{-1}}$, the given resistance to motion is $(300 + 60v)\,\mathrm{N}$, so the total opposing force is

$$600 + (300 + 60v) = (900 + 60v)\,\mathrm{N}$$

At constant speed, the resultant force is zero, so the engine's driving force must be $(900 + 60v)\,\mathrm{N}$.

With driving force F at speed v, the power is

$$P = Fv$$

so the power needed to maintain speed v is

$$P(v) = v(900 + 60v) = 900v + 60v^2$$

The power needed to maintain speed v, $v(900 + 60v)$, increases with v, so the engine can maintain every speed up to the one at which this power equals its maximum power, and no greater speed. So the greatest constant speed, $15\,\mathrm{m\,s^{-1}}$, is reached when the engine works at its maximum power.

Hence

$$P_{\max} = P(15) = 15(900 + 60 \times 15) = 15(1800) = 27\,000\,\mathrm{W}$$

So the maximum power output of the lorry's engine is $27\,000\,\mathrm{W} = 27\,\mathrm{kW}$.

Final answers with $g = 10$: $27\,200\,\mathrm{W}$

3. An airport tug of mass 1800 kg has an engine that can deliver a maximum power of 72 kW.

In an initial model of the motion of the tug, the total resistance to motion is assumed to be constant.

(i) Given that the greatest steady speed of the tug on a straight horizontal runway is $18\,\mathrm{m\,s^{-1}}$, find the magnitude of the resistance force. **[2]**

(ii) The tug is connected to a small aircraft of mass 2200 kg by a light rigid horizontal tow bar. The greatest steady speed of the tug and aircraft on the runway is now $12\,\mathrm{m\,s^{-1}}$. The resistance to motion of the aircraft may also be assumed constant.

Find the magnitude of the resistance force on the aircraft. **[2]**

(iii) The tug and aircraft again travel along the runway. At one instant their speed is $10\,\mathrm{m\,s^{-1}}$ and their acceleration is $0.25\,\mathrm{m\,s^{-2}}$.

(a) Find the power of the engine of the tug at this instant. **[4]**
(b) Find the magnitude of the force in the tow bar at this instant. **[2]**

(iv) In a refined model of the motion of the tug and aircraft, the resistance to motion of each is assumed to be zero until they reach a speed of $6\,\mathrm{m\,s^{-1}}$. When the speed is $6\,\mathrm{m\,s^{-1}}$ or above, the same constant resistance forces as in the first model are assumed to apply to each.

The tug and aircraft start from rest on the runway and accelerate, using maximum power.

Without carrying out any further calculations,

(a) explain whether the time taken to attain a speed of $10\,\mathrm{m\,s^{-1}}$ would be predicted to be lower, the same or higher using the refined model compared with the original model. **[2]**
(b) explain whether the greatest steady speed of the system would be predicted to be lower, the same or higher using the refined model compared with the original model. **[2]**

Solution

(i) At greatest steady speed on a horizontal runway, the acceleration is zero. Hence the driving force F equals the resistance R_{tug} on the tug, so $F = R_{\text{tug}}$.

Using

$$P = Fv$$

gives

$$R_{\text{tug}} = \frac{72\,000}{18} = 4000\,\mathrm{N}$$

(ii) For the tug and aircraft together, at the greatest steady speed the driving force F equals the total resistance R_{total}, so $F = R_{\text{total}}$. Hence

$$R_{\text{total}} = \frac{P}{v} = \frac{72\,000}{12} = 6000\,\mathrm{N}$$

Therefore the resistance on the aircraft is

$$R_{\text{aircraft}} = 6000 - 4000 = 2000\,\mathrm{N}$$

(iii) **(a)** The total mass is

$$1800 + 2200 = 4000\,\text{kg}$$

Take the direction of motion as positive. Using Newton's second law, the resultant force is

$$ma = 4000 \times 0.25 = 1000\,\text{N}$$

The total resistance is

$$4000 + 2000 = 6000\,\text{N}$$

Hence the driving force of the engine is

$$1000 + 6000 = 7000\,\text{N}$$

Therefore the power at this instant is

$$P = Fv = 7000 \times 10 = 70\,000\,\text{W} = 70\,\text{kW}$$

(b) Let the force in the tow bar be T.

Now consider the aircraft alone. Using Newton's second law:

$$T - 2000 = 2200 \times 0.25$$

so

$$T - 2000 = 550$$

$$T = 2550$$

So the magnitude of the force in the tow bar is 2550 N.

(iv) **(a)** The time would be *lower*.

In the refined model, there is no resistance until the speed reaches $6\,\text{m s}^{-1}$. So below $6\,\text{m s}^{-1}$, the resultant force is greater than in the original model, and hence the acceleration is greater.

From 6 to $10\,\text{m s}^{-1}$, both models are the same. Therefore the refined model predicts a smaller time to reach $10\,\text{m s}^{-1}$.

(b) The greatest steady speed would be *the same*.

From part (ii), the greatest steady speed of the system is $12\,\text{m s}^{-1}$, which is above $6\,\text{m s}^{-1}$. So at that speed, both models use the same resistance forces. Since the maximum engine power is also the same, the predicted greatest steady speed is unchanged.

4. A van of mass 1000 kg is driven up a straight road inclined at an angle α to the horizontal, where $\sin\alpha = \frac{1}{20}$. The engine is working at a constant rate of 12 kW and the van moves at a constant speed of $15\,\mathrm{m\,s^{-1}}$.

(a) Find the resistance to the motion of the van. **[4]**

(b) The van now tows a trailer of mass 600 kg up the same road using a tow rope parallel to the line of motion. The resistance to the motion of the van is the same as in part (a), and the resistance to the motion of the trailer is a constant force of magnitude 106 N.

The engine is still working at a constant rate of 12 kW.

The tow rope is modelled as light and inextensible.

Using the model, find the tension in the tow rope at the instant when the speed of the van is $7.5\,\mathrm{m\,s^{-1}}$. **[7]**

Solution

(a) Let the resistance to the motion of the van be R N.

The engine power is 12 kW = 12 000 W. Using

$$P = Fv$$

gives the driving force:

$$12\,000 = F \times 15$$

$$F = 800\,\mathrm{N}$$

The van moves at constant speed, so its acceleration is zero. Hence the resultant force along the slope is zero.

Taking up the slope as positive:

$$800 = R + 1000 \times 9.8 \times \frac{1}{20}$$

$$800 = R + 490$$

$$R = 310$$

So the resistance to the motion of the van is 310 N.

(b) When the speed is $7.5\,\mathrm{m\,s^{-1}}$, the engine is still working at 12 000 W, so the driving force is

$$F = \frac{P}{v} = \frac{12\,000}{7.5} = 1600\,\mathrm{N}$$

First consider the van and trailer together as one system.

Their total mass is

$$1000 + 600 = 1600\,\mathrm{kg}$$

The total force opposing the motion up the slope is

$$310 + 106 + 1600 \times 9.8 \times \frac{1}{20}$$

$$= 310 + 106 + 784 = 1200\,\text{N}$$

So the resultant force up the slope is

$$1600 - 1200 = 400\,\text{N}$$

Using Newton's second law:

$$a = \frac{400}{1600} = 0.25\,\text{m}\,\text{s}^{-2}$$

The tow rope is light and inextensible, so the trailer has the same acceleration.

Now consider the trailer alone. Resolving along the slope:

$$T - 106 - 600 \times 9.8 \times \frac{1}{20} = 600 \times 0.25$$

$$T - 106 - 294 = 150$$

$$T = 550$$

So the tension in the tow rope is 550 N.

Final answers with $g = 10$: (a) 300 N; (b) 554 N

5. A delivery van of mass 2000 kg is travelling up a straight road inclined at angle α to the horizontal, where $\sin\alpha = 0.05$.

The van's engine can produce power at a maximum rate of 64 kW.

The total resistance force due to friction and air resistance is modelled as $(200 + kv)$ N, where $v\,\mathrm{m\,s^{-1}}$ is the speed of the van and k is a constant.

The greatest possible constant speed of the van up the hill is $20\,\mathrm{m\,s^{-1}}$.

Find the maximum possible acceleration of the van when it is travelling uphill at $16\,\mathrm{m\,s^{-1}}$.

Fully justify your answer. **[7]**

Solution

For motion up the slope, the forces acting down the slope are the resistance and the component of weight.

The component of weight down the slope is

$$2000g\sin\alpha = 2000 \times 9.8 \times \frac{1}{20} = 980\,\mathrm{N}$$

If the engine provides driving force F when the speed is v, then

$$P = Fv$$

so

$$F = \frac{P}{v}$$

At the greatest possible constant speed, $v = 20\,\mathrm{m\,s^{-1}}$. Since the speed is constant, the acceleration is 0, so the resultant force is 0. Also, the engine must be working at its maximum power; otherwise a larger power would give a larger driving force, so the van could travel at a higher constant speed, which contradicts the statement that $20\,\mathrm{m\,s^{-1}}$ is the greatest possible constant speed.

So the driving force at $20\,\mathrm{m\,s^{-1}}$ is

$$F = \frac{64\,000}{20} = 3200\,\mathrm{N}$$

The resistance at this speed is

$$200 + 20k$$

Balancing forces along the slope gives

$$3200 = (200 + 20k) + 980$$

$$3200 = 1180 + 20k$$

$$20k = 2020$$

$$k = 101$$

Now consider the van when it is travelling at $16\,\mathrm{m\,s^{-1}}$.

To get the maximum possible acceleration, we need the maximum possible resultant force up the slope. At this speed, the weight component and the resistance are fixed, so this happens when the engine again uses its maximum power.

Hence the maximum possible driving force is

$$F = \frac{64\,000}{16} = 4000\,\text{N}$$

The resistance force at $16\,\text{m}\,\text{s}^{-1}$ is

$$200 + kv = 200 + 101(16) = 1816\,\text{N}$$

So the resultant force up the slope is

$$4000 - 1816 - 980 = 1204\,\text{N}$$

Take up the slope as positive. Using Newton's second law:

$$1204 = 2000a$$

so

$$a = \frac{1204}{2000} = 0.602\,\text{m}\,\text{s}^{-2}$$

Final answers with $g = 10$: $0.6\,\text{m}\,\text{s}^{-2}$

6. A delivery van has mass 1300 kg and travels on a straight horizontal road.

In a simple model, the van experiences a constant rolling resistance of 600 N and an additional air-resistance force whose magnitude is proportional to the speed of the van.

During a test, the van moves at a steady speed of $15\,\mathrm{m\,s^{-1}}$ when the engine provides a driving force of 1200 N.

In this model, the engine can deliver a constant maximum power. The greatest speed the van can attain on this road is $30\,\mathrm{m\,s^{-1}}$.

(a) Show that the maximum power of the engine is 54 000 W. **[4]**

(b) Find the maximum acceleration of the van when it is travelling at $20\,\mathrm{m\,s^{-1}}$. **[2]**

(c) Find the maximum acceleration of the van when it is travelling at $4\,\mathrm{m\,s^{-1}}$. **[2]**

(d) Comment on the validity of the model in the context of your answers to parts (b) and (c). **[2]**

Solution

(a) Let the air-resistance force be kv, where v is the speed.

At a steady speed of $15\,\mathrm{m\,s^{-1}}$, the acceleration is zero, so the driving force equals the total resistance:

$$1200 = 600 + 15k$$

Hence

$$15k = 600 \quad \Longrightarrow \quad k = 40$$

So the total resistance at speed v is

$$600 + 40v$$

At the greatest speed $30\,\mathrm{m\,s^{-1}}$, the van is again moving steadily, so the driving force F is

$$F = 600 + 40(30) = 1800\,\mathrm{N}$$

With driving force F at speed v, the power is

$$P = Fv$$

so

$$P = 1800 \times 30 = 54\,000\,\mathrm{W}$$

as required.

(b) At $20\,\mathrm{m\,s^{-1}}$, using the maximum power found in part (a):

$$F = \frac{P}{v} = \frac{54\,000}{20} = 2700\,\mathrm{N}$$

So the maximum driving force is 2700 N.

The resistance at $20\,\text{m s}^{-1}$ is

$$600 + 40(20) = 1400\,\text{N}$$

Therefore the resultant forward force is

$$2700 - 1400 = 1300\,\text{N}$$

Take the direction of motion as positive. Using Newton's second law:

$$1300 = 1300a$$

so

$$a = 1\,\text{m s}^{-2}$$

(c) At $4\,\text{m s}^{-1}$, the maximum driving force is

$$F = \frac{54\,000}{4} = 13\,500\,\text{N}$$

The resistance is

$$600 + 40(4) = 760\,\text{N}$$

So the resultant force is

$$13\,500 - 760 = 12\,740\,\text{N}$$

Using Newton's second law:

$$a = \frac{12\,740}{1300} = 9.8\,\text{m s}^{-2}$$

(d) The answer in part (b), $1\,\text{m s}^{-2}$, is reasonable for a delivery van.

However, the answer in part (c), $9.8\,\text{m s}^{-2}$, is unrealistically large for a van. This suggests the model is not valid at low speeds.

The reason is that, with constant power, the driving force is

$$F = \frac{P}{v}$$

so as v gets smaller, the model predicts a much larger driving force. In fact, as $v \to 0$, the force would become extremely large, which is not realistic.

So the constant-power model is reasonable at higher speeds, but not at low speeds.

7. A maintenance vehicle of mass 1000 kg moves on a straight slope inclined at an angle θ to the horizontal, where $\sin\theta = 0.08$. When the motor is switched off, the vehicle coasts down the slope and reaches a limiting speed of $14\,\mathrm{m\,s^{-1}}$. The resistance to motion is proportional to the square of the speed. The motor is then switched on and works at a constant power P W, allowing the vehicle to travel up the same slope with a limiting speed of $10\,\mathrm{m\,s^{-1}}$.

Find the value of P and determine the resistance to motion when the vehicle is travelling at $12\,\mathrm{m\,s^{-1}}$. **[7]**

Solution

Let the resistance to motion at speed v be

$$R = kv^2$$

where k is a constant.

At a limiting speed, the acceleration is zero, so the resultant force along the slope is zero.

When the motor is switched off, the vehicle moves *down* the slope with limiting speed $14\,\mathrm{m\,s^{-1}}$. The component of the weight down the slope is

$$mg\sin\theta = 1000 \times 9.8 \times 0.08 = 784\,\mathrm{N}$$

At limiting speed, this is balanced by the resistance, so

$$k(14)^2 = 784$$

Hence

$$196k = 784$$

so

$$k = 4$$

Therefore the resistance is

$$R = 4v^2$$

Now the motor is switched on and the vehicle travels *up* the slope with limiting speed $10\,\mathrm{m\,s^{-1}}$.

With driving force F at speed v, the power is

$$P = Fv$$

So when $v = 10$:

$$F = \frac{P}{10}$$

The forces opposing the upward motion are

$$\text{weight component} = 784\,\mathrm{N}$$

and

$$\text{resistance} = 4(10)^2 = 400\,\mathrm{N}$$

Since the speed is limiting, resultant force is zero, so

$$\frac{P}{10} = 784 + 400$$

Hence

$$\frac{P}{10} = 1184$$

and therefore

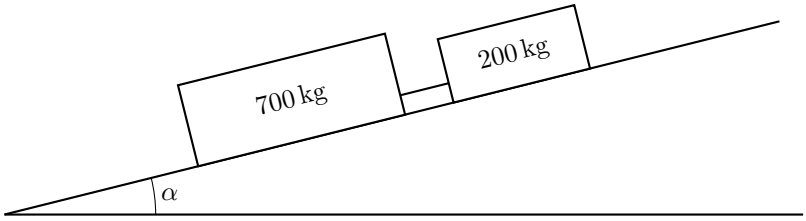
$$P = 11\,840$$

When the vehicle is travelling at $12\,\mathrm{m\,s^{-1}}$, the resistance is

$$R = 4(12)^2 = 4 \times 144 = 576\,\mathrm{N}$$

Therefore, the motor power is $11\,840\,\mathrm{W}$ ($11.84\,\mathrm{kW}$), and the resistance to motion at $12\,\mathrm{m\,s^{-1}}$ is $576\,\mathrm{N}$.

Final answers with $g = 10$: $P = 12\,100$; resistance $588\,\mathrm{N}$



8. A car of mass 700 kg is descending a straight road which is inclined at an angle α to the horizontal, where $\sin\alpha = \frac{1}{10}$. The car is connected to a trailer of mass 200 kg by a light rigid towbar which is parallel to the road, as shown in the diagram.

The resistance to the motion of the car is modelled as a constant force of magnitude 240 N.
The resistance to the motion of the trailer is modelled as a constant force of magnitude 90 N.

The brakes of the car dissipate energy at a constant rate of 9 kW.

Find the force in the towbar at the instant when the speed of the car is $12\,\text{m}\,\text{s}^{-1}$, stating whether the towbar is in tension or compression. **[8]**

Solution

Since the car is moving *down* the slope, the resistances and the braking force all act *up* the slope.

The brakes dissipate energy at a rate of 9 kW = 9000 W. Using

$$P = Fv$$

at the instant when $v = 12\,\text{m}\,\text{s}^{-1}$, the braking force is

$$F = \frac{P}{v} = \frac{9000}{12} = 750\,\text{N}$$

This force acts up the slope.

Take up the slope as positive.

For the car and trailer together, the total mass is

$$700 + 200 = 900\,\text{kg}$$

The component of their total weight down the slope is

$$900g\sin\alpha = 900g\left(\frac{1}{10}\right) = 90g$$

So, resolving along the slope for the whole system:

$$240 + 90 + 750 - 90g = 900a$$

$$1080 - 90g = 900a$$

With $g = 9.8$:

$$1080 - 90(9.8) = 900a$$

$$1080 - 882 = 900a$$

$$198 = 900a$$

$$a = \frac{198}{900} = 0.22\,\mathrm{m\,s^{-2}}$$

So the acceleration is $0.22\,\mathrm{m\,s^{-2}}$ up the slope.

Now consider the trailer alone. Let T be the force in the towbar on the trailer, taken positive up the slope.

For the trailer, resolving along the slope:

$$T + 90 - 200g\sin\alpha = 200a$$

$$T + 90 - 200g\left(\frac{1}{10}\right) = 200(0.22)$$

$$T + 90 - 20g = 44$$

Substitute $g = 9.8$:

$$T + 90 - 196 = 44$$

$$T - 106 = 44$$

$$T = 150$$

Hence the force in the towbar has magnitude

$$150\,\mathrm{N}$$

This force on the trailer is *up* the slope. If the towbar were in tension, it would pull the trailer *down* the slope towards the car. So the towbar must be in **compression**.

Therefore, the force in the towbar is 150 N, in compression.

9. A snowmobile of mass 800 kg tows a rescue sledge of mass 200 kg up a straight slope that is inclined at angle θ to the horizontal. The sledge is attached to the snowmobile by a light inextensible rope. The resistance to the motion of the snowmobile from non-gravitational forces is modelled as a constant force of magnitude 216 N. The resistance to the motion of the sledge from non-gravitational forces is modelled as a constant force of magnitude 294 N.

When the engine of the snowmobile is working at a constant rate of 12 kW, the snowmobile and the sledge move up the slope with constant speed $12\,\mathrm{m\,s^{-1}}$.

(a) Find the value of $\sin\theta$. **[4]**

(b) Later, at the instant when the snowmobile and the sledge are travelling down the slope with speed $14\,\mathrm{m\,s^{-1}}$, the rope snaps. When the rope snaps the sledge is at the point A. The sledge continues to travel down the slope before coming to instantaneous rest at the point B. The resistance to the motion of the sledge from non-gravitational forces is again modelled as a constant force of magnitude 294 N.

Use the work–energy principle, and your answer to part (a), to find the distance AB. **[5]**

Solution

(a) Take the snowmobile and sledge together as one system, so the rope tension is internal.

With driving force F at speed v, the power is

$$P = Fv$$

so

$$\begin{aligned} F &= \frac{P}{v} \\ &= \frac{12\,000}{12} \\ &= 1000\,\mathrm{N} \end{aligned}$$

The total mass is

$$800 + 200 = 1000\,\mathrm{kg}$$

So the component of the total weight down the slope is $1000g\sin\theta$, and the total resistance is

$$216 + 294 = 510\,\mathrm{N}$$

As the speed is constant, the resultant force along the slope is zero. Hence

$$1000 = 1000g\sin\theta + 510$$

So

$$\begin{aligned} 1000g\sin\theta &= 490 \\ 9800\sin\theta &= 490 \\ \sin\theta &= 0.05 \end{aligned}$$

(b) Let $AB = s$ m. The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Take state 0 at A and state 1 at B.

For the sledge:

$$KE_0 = \tfrac{1}{2} \times 200 \times 14^2 = 19\,600 \text{ J}$$
$$KE_1 = 0$$

The only other force doing work is the resistance, which acts up the slope while the sledge moves down the slope, so

$$\sum W_{\text{other}} = -294s$$

Take B as the zero level for gravitational potential energy. The vertical drop from A to B is $s\sin\theta$, so

$$GPE_0 = 200gs\sin\theta, \qquad GPE_1 = 0$$

Substituting into the work–energy equation:

$$19\,600 + 200gs\sin\theta - 294s = 0$$

Using $\sin\theta = 0.05$:

$$19\,600 + 200(9.8)(0.05)s - 294s = 0$$
$$19\,600 + 98s - 294s = 0$$
$$19\,600 - 196s = 0$$
$$s = 100$$

Therefore

$$AB = 100 \text{ m}$$

Final answers with $g = 10$: (a) $\sin\theta = 0.049$

10. A lorry of mass 3200 kg tows a trailer of mass 1800 kg down a line of greatest slope of a straight road inclined at an angle θ to the horizontal, where $\sin\theta = 0.06$.

The power developed by the lorry's engine is constant. The resistances to motion of the lorry and the trailer are constant and equal to 800 N and 500 N respectively. When the lorry passes through a point A on the road, the speed of the combination is $14\,\mathrm{m\,s^{-1}}$ and its acceleration is $1.2\,\mathrm{m\,s^{-2}}$.

(a) Calculate the power developed by the engine. **[4]**

(b) The lorry later passes through a point B on the road with speed $22\,\mathrm{m\,s^{-1}}$. The time taken for the lorry to travel from A to B is 7.93 s.

Find the distance AB. **[5]**

Solution

(a) The total mass of the lorry and trailer is

$$3200 + 1800 = 5000\,\mathrm{kg}$$

Take the direction down the slope as positive.

With driving force F N at speed $v\,\mathrm{m\,s^{-1}}$, the power is

$$P = Fv$$

Resolving forces down the slope for the lorry-trailer combination at A:

$$F + 5000g\sin\theta - (800 + 500) = 5000(1.2)$$

Substitute $g = 9.8$ and $\sin\theta = 0.06$:

$$F + 5000(9.8)(0.06) - 1300 = 6000$$

$$F + 2940 - 1300 = 6000$$

$$F = 4360$$

Hence

$$P = Fv = 4360 \times 14 = 61\,040\,\mathrm{W}$$

So the power developed by the engine is

$$61.04\,\mathrm{kW}$$

(b) Let the distance AB be s m.

The work–energy principle is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Take state 0 at A and state 1 at B. Take B as the zero level for gravitational potential energy.

Here:

- the other forces doing work are the engine's driving force, doing work Pt from A to B, and the resistances, doing work $-(800+500)s = -1300s$, so $\sum W_{\text{other}} = Pt - 1300s$,
- A is $s\sin\theta$ above B, so

$$GPE_0 = 5000g(s\sin\theta) = 5000 \times 9.8 \times 0.06\, s = 2940s$$

and $GPE_1 = 0$.

Therefore

$$\frac{1}{2}(5000)(14^2) + 2940s + Pt - 1300s = \frac{1}{2}(5000)(22^2)$$

With $P = 61\,040\,\text{W}$ and $t = 7.93\,\text{s}$:

$$\frac{1}{2}(5000)(14^2) + 2940s + 61\,040(7.93) - 1300s = \frac{1}{2}(5000)(22^2)$$

$$61\,040(7.93) + 1640s = \frac{1}{2}(5000)(22^2 - 14^2)$$

$$61\,040(7.93) + 1640s = 720\,000$$

So

$$s = \frac{720\,000 - 61\,040(7.93)}{1640} \approx 143.9$$

Hence

$$AB \approx 144\,\text{m}$$

Final answers with $g = 10$: (a) 60.2 kW; (b) 143 m

11. A lorry of mass 2500 kg travels up a straight road inclined at an angle α to the horizontal, where $\tan\alpha = \frac{7}{24}$.

The engine of the lorry supplies a constant power of P W. The total resistance to motion is a constant force of magnitude R N.

When the lorry is moving at $12\,\mathrm{m\,s^{-1}}$, its acceleration is $2.4\,\mathrm{m\,s^{-2}}$. When it is moving at $18\,\mathrm{m\,s^{-1}}$, its acceleration is $0.4\,\mathrm{m\,s^{-2}}$.

Find the value of P and the value of R. **[8]**

Solution

Take motion up the slope as positive.

Since $\tan\alpha = \frac{7}{24}$, we use the $7:24:25$ triangle, so

$$\sin\alpha = \frac{7}{25}$$

Hence the component of the lorry's weight acting down the slope is

$$mg\sin\alpha = 2500 \times 9.8 \times \frac{7}{25} = 6860\,\mathrm{N}$$

With driving force F at speed v, the power is

$$P = Fv$$

so

$$F = \frac{P}{v}$$

So, resolving along the slope:

$$\frac{P}{v} - R - 6860 = 2500a$$

Using $v = 12$ and $a = 2.4$:

$$\frac{P}{12} - R - 6860 = 2500(2.4)$$

$$\frac{P}{12} - R - 6860 = 6000$$

$$\frac{P}{12} - R = 12\,860$$

Using $v = 18$ and $a = 0.4$:

$$\frac{P}{18} - R - 6860 = 2500(0.4)$$

$$\frac{P}{18} - R - 6860 = 1000$$

$$\frac{P}{18} - R = 7860$$

Now subtract the second equation from the first:

$$\left(\frac{P}{12} - R\right) - \left(\frac{P}{18} - R\right) = 12\,860 - 7860$$

$$\frac{P}{12} - \frac{P}{18} = 5000$$

$$P\left(\frac{1}{12} - \frac{1}{18}\right) = 5000$$

$$P\left(\frac{1}{36}\right) = 5000$$

So

$$P = 180\,000$$

Substitute into $\frac{P}{18} - R = 7860$:

$$\frac{180\,000}{18} - R = 7860$$

$$10\,000 - R = 7860$$

$$R = 2140$$

Therefore

$$P = 180\,000, \qquad R = 2140$$

Final answers with $g = 10$: $R = 2000$

12. A rail maintenance vehicle of mass 2000 kg travels in a straight line along a horizontal section of track at a constant speed of $20\,\mathrm{m\,s^{-1}}$.

The resistance to the motion of the vehicle is modelled as a constant force of magnitude 2000 N.

The engine is working at a constant rate of P kW.

Using the model,

(a) find the value of P. **[2]**

The vehicle now travels up a straight section of track which is inclined to the horizontal at an angle α, where $\sin\alpha = \frac{1}{20}$.

In a refined model, the resistance to the motion of the vehicle from non-gravitational forces is modelled as a force of magnitude $(220 + 2v^2)$ N, where $v\,\mathrm{m\,s^{-1}}$ is the speed.

At the instant when the engine is working at the same constant rate as in part (a) and the vehicle is moving up the track at $10\,\mathrm{m\,s^{-1}}$, the acceleration of the vehicle is $a\,\mathrm{m\,s^{-2}}$.

Using the refined model,

(b) find the value of a. **[4]**

Later, when the engine is again working at this constant rate, the vehicle is moving up the track at a constant speed $U\,\mathrm{m\,s^{-1}}$.

Using the refined model,

(c) find the value of U. **[5]**

Solution

(a) On the horizontal track the vehicle moves at constant speed, so the resultant force is zero. Hence the driving force is equal to the resistance:

$$F = 2000\,\mathrm{N}$$

The engine works at P kW, which is $1000P$ W. With driving force F at speed v, the power is

$$1000P = Fv$$

so

$$1000P = 2000 \times 20 = 40\,000$$

$$P = 40$$

(b) From part (a), the engine power is constant at

$$40\,000\,\mathrm{W}$$

When the speed is $10\,\mathrm{m\,s^{-1}}$, the driving force is

$$F = \frac{40\,000}{10} = 4000\,\mathrm{N}$$

The non-gravitational resistance is

$$220 + 2(10)^2 = 220 + 200 = 420\,\text{N}$$

The component of the weight down the slope is

$$mg\sin\alpha = 2000 \times 9.8 \times \frac{1}{20} = 980\,\text{N}$$

So the total force opposing the motion is

$$420 + 980 = 1400\,\text{N}$$

Hence the resultant force up the slope is

$$4000 - 1400 = 2600\,\text{N}$$

Take up the slope as positive. Using Newton's second law:

$$ma = 2600$$

so

$$2000a = 2600$$

$$a = 1.3$$

(c) When the vehicle is moving at constant speed U, its acceleration is zero, so the resultant force along the slope is zero.

The driving force is found from the constant power:

$$F = \frac{40\,000}{U}$$

The forces opposing the motion are:

- non-gravitational resistance: $\left(220 + 2U^2\right)\text{N}$
- component of weight down the slope: $2000 \times 9.8 \times \frac{1}{20} = 980\,\text{N}$

So

$$\frac{40\,000}{U} = (220 + 2U^2) + 980$$

which simplifies to

$$\frac{40\,000}{U} = 1200 + 2U^2$$

Multiplying by U:

$$40\,000 = 1200U + 2U^3$$

$$U^3 + 600U - 20\,000 = 0$$

Try $U = 20$:

$$20^3 + 600(20) - 20\,000 = 8000 + 12\,000 - 20\,000 = 0$$

So $(U - 20)$ is a factor. Hence

$$U^3 + 600U - 20\,000 = (U - 20)(U^2 + 20U + 1000)$$

The discriminant of the quadratic factor is

$$b^2 - 4ac = 20^2 - 4(1)(1000) = 400 - 4000 = -3600 < 0$$

so it has no real roots.

Therefore the only real value is

$$U = 20$$

Final answers with $g = 10$: (b) $a = 1.29$; (c) $U = 19.9$